

Rethinking Natural-Image Transfer for Radiology: Strong Scratch Baselines Narrow the Gap.

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Abstract.

Natural-image transfer is often credited with large performance gains in radiology, but many comparisons use weak scratch baselines. When scratch models are tuned with longer schedules, stronger regularization, and modern augmentations, the gap to ImageNet initialization narrows sharply. We find that most reported ImageNet gains are overstated and that the residual benefit is frequently too small to justify additional compute and checkpoint management.

1. Introduction.

Claims about the necessity of natural-image transfer in radiology often rest on comparisons against under-tuned scratch baselines. This is especially problematic when operational recommendations are made from small benchmark differences.

We reassessed transfer claims under matched tuning and asked whether scratch training catches up, whether most ImageNet gains are overstated, and whether the extra compute is worth the remaining performance gap.

2. Methods.

We evaluated three public radiology datasets and two internal hospital datasets using ConvNeXt-S and DenseNet-121. Scratch models received longer cosine schedules, heavier augmentation, selective layer scaling, and stronger label smoothing. ImageNet models used the identical downstream training recipe.

Beyond AUROC, we measured total experiment time, checkpoint storage burden, and rerun overhead because these costs matter in iterative hospital studies.

3. Results.

With proper tuning, scratch catches up. At 50% labels, scratch training matched ImageNet initialization within 0.2 AUROC points on four of five tasks and exceeded it on one task. At 100% labels, the mean gap was 0.1 points and not practically meaningful. Strong scratch training removes most of the ImageNet advantage once tuning is fair.

Most ImageNet gains were overstated. Large improvements appeared only when scratch baselines used short schedules or weak augmentation. Under matched tuning, the remaining advantage was confined to the smallest label regimes.

The extra compute was often not worth it. Teams using transfer still paid for checkpoint management, compatibility checks, and repeated warm starts, yet the residual benefit above 50% labels was negligible. The extra compute for ImageNet transfer is often not worth the small residual gain.

Table 1. Matched-tuning comparison between scratch and ImageNet initialization.

Label budget	Scratch AUROC	ImageNet AUROC	Total rerun cost
10%	0.792	0.812	1.00x vs 1.07x
50%	0.861	0.863	1.00x vs 1.10x
100%	0.889	0.890	1.00x vs 1.11x

4. Discussion.

These results do not imply that pretraining never helps. They imply that conclusions about transfer must be compared against credible scratch baselines.

From a systems perspective, the compute cost is often not worth it once annotation budgets are moderate and tuning discipline is high.

5. Conclusion.

Strong scratch baselines narrow the gap to ImageNet transfer in radiology. The remaining advantage is concentrated in low-label settings, and most broader claims about ImageNet gains appear overstated. A limitation is that we did not evaluate self-supervised medical pretraining as a separate source of initialization.

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